Run report: PKM Kummer tower, m = 3, N = 21 with msolve, run 20261018T0627Z

Study `research/pkm_m3_n21_msolve_20261018/`, pod `zdec0w9xwevehz` (8 vCPU, cgroup 32 GB, cap 27.2 GB, one thread), commit `6667c066`, msolve 0.6.5 (Ubuntu package, sha256 `c2722288…`). Observations only; the decision-rule reading is in `RESULTS.md`.

Runs

system	N	role	wall	outcome	rounds	max deg	peak VmHWM
`m3_t5_x1289903598`	15	control, committed D = 7	2,700 s	finished, exit 0, 1,299 s, one-element Gröbner basis	25	7	2.35 GB
`m3_t7_x221026520`	21	target 0	15,458 s	wall stop inside the 6th degree-7 round	25	7	15.33 GB
`m3_t7_x704078485`	21	target 1	7,728 s	wall stop inside the 4th degree-7 round	23	7	8.88 GB
`m3_t8_x1227422598`	24	extra	8,328 s	wall stop inside the 5th degree-7 round	24	7	13.03 GB

Rounds and degrees from msolve's `-v 2` round table; VmHWM sampled once a second from `/proc` (measured).

Degree-7 rounds at N = 21, target 0

round	pairs	matrix	density	new	zero	round s
1	20,631	98,282 × 101,435	1.06 %	2,354	18,321	154.36
2	29,876	170,445 × 194,842	1.22 %	4,269	25,607	1,127.36
3	33,319	203,174 × 228,589	1.45 %	5,421	28,044	2,498.79
4	28,015	255,388 × 307,477	1.45 %	5,043	23,096	4,633.73
5	20,804	323,437 × 431,738	1.06 %	5,502	15,486	5,812.61
6	7,038	384,152 × 528,099	0.75 %	stopped		

Every completed degree-7 round produced thousands of new basis elements; no run reached the end of degree 7 and none selected a degree-8 pair. The N = 15 control finished with maximum degree 7 and a one-element Gröbner basis, as item 3 recorded for the same system (committed D = 7).

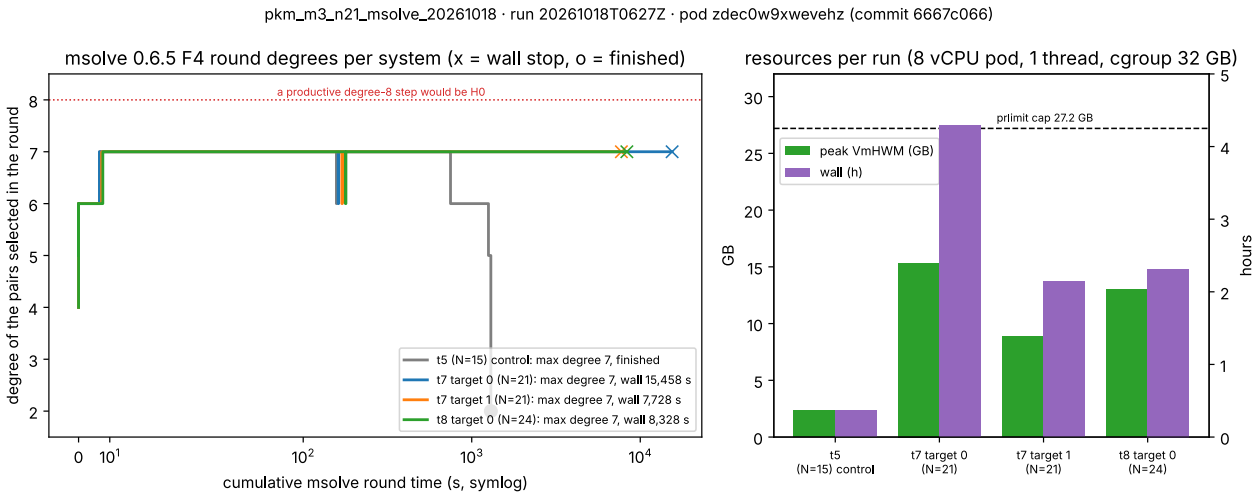


Figure 1: Run 20261018T0627Z, pod `zdec0w9xwevehz`. Left: degree of the pairs selected in each msolve round against cumulative round time per system (symlog), stop marked. Right: peak VmHWM and wall per run against the 27.2 GB cap. Recorded values only (`plot.py`).

Missing

No N = 21 or N = 24 run finished, so no D is recorded for them; `m3_t8` target 1 was not run; the `f4_fp_tower` cell stays blocked on a 192 GB host. Raw records in `runs/` and `progress.txt`.